

LAB 1 — USE-CASE FLOW SPECIFICATION

Problem Statement: Blood Bank Inventory & Emergency Donor Matcher

Use Case: UC-01 — Submit Emergency Blood Request

Preconditions

1. The Emergency Requester is authorized to submit a blood request.
2. The blood bank system is available.
3. Blood group and required unit information can be entered.

Postconditions

1. The emergency request is stored with a unique request ID.
2. The request status is updated based on blood availability.
3. If blood is issued, the blood inventory is updated.
4. If blood is unavailable, suitable donors are searched and notified.

Main Success Scenario

1. Emergency Requester selects the "Submit Emergency Blood Request" option.
2. System displays a blood request form.
3. Emergency Requester enters the required blood group and number of units.
4. Emergency Requester submits the request.
5. System validates the entered information.
6. System checks the available blood inventory.
7. System finds that the required blood group and quantity are available.
8. System displays the available blood units to the Emergency Requester.
9. Blood Bank Manager processes the blood request.
10. System updates the blood inventory after the blood is issued.
11. System displays a confirmation message and request status.
12. Use case ends successfully.

Alternate Flow — Required Blood Not Available

- 6a1. System finds that the required blood group or quantity is not available.
- 6a2. System searches for suitable available donors.
- 6a3. System displays matching donors.
- 6a4. System sends an emergency request notification to suitable donors.
- 6a5. If a donor accepts the request, the Blood Bank Manager coordinates the donation.
- 6a6. System records the donation and updates the blood inventory.
- 6a7. Blood Bank Manager processes the emergency request.
- 6a8. System updates the request status.